MIT | Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

PRACTICAL NO.02		6 HOURS

Practice Lab Assignment:

1. Perform all List Operations, Dictionary Operations.

Lab Assignment:

1. Select the number from the entered list and find its position in Python (use Linear Search).
2. Choose cricket team of eleven players find the captain of the team (consider tallest person as a captain) using dictionary.

Practice Lab Assgmt 02

List Operations

```
# List operations demo

lst = []

n = int(input("Enter number of elements: "))

for i in range(n):
    lst.append(int(input("Enter element: ")))

print("Original List:", lst)

# Operations
lst.append(100)
print("After append:", lst)

lst.insert(1, 50)
print("After insert:", lst)

lst.remove(lst[0])
print("After remove first element:", lst)

print("Sorted List:", sorted(lst))

print("Max =", max(lst))
print("Min =", min(lst))
```

Output

```
Enter number of elements: 4
Enter element: 21
Enter element: 76
Enter element: 78
Enter element: 423
Original List: [21, 76, 78, 423]
After append: [21, 76, 78, 423, 100]
After insert: [21, 50, 76, 78, 423, 100]
After remove first element: [50, 76, 78, 423, 100]
Sorted List: [50, 76, 78, 100, 423]
Max = 423
Min = 50
```

Dictionary Operations

```
# Dictionary operations demo

student = {}

n = int(input("Enter number of entries: "))

for i in range(n):
    key = input("Enter name: ")
    value = int(input("Enter marks: "))
    student[key] = value

print("Dictionary:", student)

# Operations
print("Keys:", student.keys())
print("Values:", student.values())

# Update
name = input("Enter name to update marks: ")
student[name] = int(input("Enter new marks: "))

print("Updated Dictionary:", student)
```

Output

```
Enter number of entries: 5
Enter name: Bhavin
Enter marks: 76
Enter name: Kartik
Enter marks: 21
Enter name: om
Enter marks: 45
Enter name: jai
Enter marks: 87
Enter name: Karan
Enter marks: 67
Dictionary: {'Bhavin': 76, 'Kartik': 21, 'om': 45, 'jai': 87, 'Karan': 67}
Keys: dict_keys(['Bhavin', 'Kartik', 'om', 'jai', 'Karan'])
Values: dict_values([76, 21, 45, 87, 67])
Enter name to update marks: manaav
Enter new marks: 87
Updated Dictionary: {'Bhavin': 76, 'Kartik': 21, 'om': 45, 'jai': 87, 'Karan': 67, 'manaav': 87}
```

Lab Assgmt1: Linear Search(Find Position)

```
lst = []

n = int(input("Enter size of list: "))

for i in range(n):
    lst.append(int(input("Enter element: ")))

key = int(input("Enter number to search: "))

found = False

for i in range(len(lst)):
    if lst[i] == key:
        print("Element found at position:", i + 1)
        found = True
        break

if not found:
    print("Element not found")
```

Output

```
Enter size of list: 2
Enter element: 2
Enter element: 31
Enter number to search: 2
Element found at position: 1
```

Lab Assgmt: Cricket Team(Dictionary+Clg Captain)

```python
team = {}

n = 11

print("Enter player name and height:")

for i in range(n):
    name = input("Player name: ")
    height = float(input("Height (in cm): "))
    team[name] = height

print("\nTeam Data:", team)

# Find tallest player
captain = max(team, key=team.get)

print("Captain (Tallest Player):", captain)
print("Height:", team[captain])
```

Output

```
Enter player name and height:
Player name: root
Height (in cm): 194
Player name: dhoni
Height (in cm): 182
Player name: jert
Height (in cm): 173
Player name: hgu
Height (in cm): 4
Player name: edt
Height (in cm): 56
Player name: rtet
Height (in cm): 75
Player name: drtd
Height (in cm): 567
Player name: tydey
Height (in cm): 67
Player name: dfws
Height (in cm): 43
Player name: swfrwr
Height (in cm): 24
Player name: gffeg
Height (in cm): 65

Team Data: {'root': 194.0, 'dhoni': 182.0, 'jert': 173.0, 'hgu': 4.0, 'edt': 56.0, 'rtet': 75.0, 'drtd': 567.0,
'tydey': 67.0, 'dfws': 43.0, 'swfrwr': 24.0, 'gffeg': 65.0}
Captain (Tallest Player): drtd
Height: 567.0
```